E1 Solutions: differential equations and dynamical systems

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Numerical answers were checked with Python/SymPy.

Practice

1. (a) $u_1 = 5$, $u_2 = 9$, $u_3 = 17$, $u_4 = 33$.

(b) $L = 2L - 1$ gives $L = 1$. With $a = 2$: $u_n = 1 + 2^n(3-1) = 1 + 2^{n+1}$. Check: $u_4 = 1 + 32 = 33$. ✓ The fixed point is unstable ($|a| = 2 > 1$), so the sequence runs away from it.

2. $u_{10} - u_3 = 7v = 21$, so $v = 3$ and $u_0 = u_3 - 3v = 2$. Then $u_{20} = 2 + 60 = 62$ and the sum of the 21 terms is

$$\tfrac{21}{2}(u_0 + u_{20}) = \tfrac{21}{2} \cdot 64 = 672.$$

3. (a) $L = 0.6L + 8 \Rightarrow 0.4L = 8 \Rightarrow L = 20$.

(b) $u_n = 20 + 0.6^n(5 - 20) = 20 - 15 \cdot 0.6^n$. Check: $u_1 = 0.6 \cdot 5 + 8 = 11 = 20 - 9$. ✓

(c) Since $0 < 0.6 < 1$, $0.6^n \to 0$ monotonically: $u_n \to 20$ monotonically (increasing, always below 20).

4. (a) Overnight 30% is broken down, so $0.7u_n$ remains; then the new dose of 50 mg is added: $u_{n+1} = 0.7u_n + 50$.

(b) $L = 50/0.3 = 166\frac{2}{3}$ mg; it is stable since $0 < 0.7 < 1$.

(c) $u_n = 166\frac{2}{3} + 0.7^n(50 - 166\frac{2}{3}) = 166\frac{2}{3} - 116\frac{2}{3} \cdot 0.7^n$. We need

$$116\tfrac{2}{3} \cdot 0.7^n < 6\tfrac{2}{3} \iff 0.7^n < \tfrac{20}{350} = \tfrac{2}{35} \iff n > \frac{\ln(2/35)}{\ln 0.7} \approx 8.02.$$

(The inequality flips because $\ln 0.7 < 0$.) So from day $n = 9$ on. Check: $u_8 \approx 159.95$, $u_9 \approx 161.96$.

5. (a) $u_1 = 3$, $u_2 = 1.5$, $u_3 = 2.25$, $u_4 = 1.875$.

(b) The graph of $f(x) = -0.5x + 3$ is a line with slope -0.5; the web diagram is a rectangular spiral closing in on the intersection with $y = x$. Behaviour: oscillating convergence.

(c) $L = -0.5L + 3 \Rightarrow L = 2$. Indeed $u_n = 2 - 2(-0.5)^n \to 2$.

6. Left: $y' = 6e^{2t}$. Right: $2(3e^{2t} - 1) + 2 = 6e^{2t}$. Equal for all t, so it is a solution. $y(0) = 3 - 1 = 2$.

7. (a) Slope c where $x - y = c$, i.e. on the line $y = x - c$. So the isoclines are lines of slope 1: slope 0 on $y = x$, slope 1 on $y = x - 1$, slope -1 on $y = x + 1$, slope 2 on $y = x - 2$, and so on. Above $y = x$ slopes are negative, below positive.

(b) $y = x - 1$: left $y' = 1$; right $x - (x - 1) = 1$. ✓

(c) $y = x - 1 + Ce^{-x}$: left $y' = 1 - Ce^{-x}$; right $x - (x - 1 + Ce^{-x}) = 1 - Ce^{-x}$. ✓ As $x \to \infty$, $Ce^{-x} \to 0$, so every solution approaches the line $y = x - 1$ (an oblique asymptote, *scheve asymptoot*).

8. Constant solution $y = 0$ does not satisfy $y(0) = 3$. For $y \neq 0$:

$$\int \frac{dy}{y} = \int 2x\,dx \Rightarrow \ln|y| = x^2 + C \Rightarrow y = Ae^{x^2}.$$

$y(0) = A = 3$, so $y = 3e^{x^2}$. Check: $y' = 6xe^{x^2} = 2x \cdot 3e^{x^2}$. ✓

9. $\int y\,dy = \int x\,dx \Rightarrow \frac{1}{2}y^2 = \frac{1}{2}x^2 + C$. With $y(0) = 2$: $C = 2$, so $y^2 - x^2 = 4$. Since $y(0) > 0$ we take $y = \sqrt{x^2 + 4}$, defined for all x. The solution curve is the upper branch of the hyperbola $y^2 - x^2 = 4$ (you will meet hyperbolas in E3).

10. Euler: $y_{k+1} = y_k + 0.5(t_k + y_k)$.

- $t_0 = 0$, $y_0 = 1$: slope 1, $y_1 = 1.5$.
- $t_1 = 0.5$, $y_1 = 1.5$: slope 2, $y_2 = 2.5$.

So $y(1) \approx 2.5$. Check the exact solution: $y = 2e^t - t - 1$ gives $y' = 2e^t - 1$ and $t + y = 2e^t - 1$ ✓, and $y(0) = 2 - 0 - 1 = 1$ ✓. Exact $y(1) = 2e - 2 \approx 3.437$. The error, 0.94, is large because h is large and the solution is convex (Euler follows tangent lines, which lie below a convex curve).

Exam-level

11. (a) Put $D = T - 20$. Then $D' = T' = -kD$, so $D = D_0 e^{-kt}$ with $D_0 = 70$: $T(t) = 20 + 70e^{-kt}$.

(b) $T(5) = 70$: $70e^{-5k} = 50 \Rightarrow e^{-5k} = \frac{5}{7} \Rightarrow k = \frac{1}{5}\ln\frac{7}{5}$.

(c) $20 + 70e^{-kt} = 40 \Rightarrow e^{-kt} = \frac{2}{7} \Rightarrow t = \dfrac{\ln 3.5}{k} = \dfrac{5\ln 3.5}{\ln 1.4} \approx 18.6$ minutes.

12. (a) $N = 0$ and $N = 1000$ are constant solutions (not ours). Otherwise, with partial fractions $\dfrac{1000}{N(1000-N)} = \dfrac{1}{N} + \dfrac{1}{1000-N}$:

$$\int \left(\frac{1}{N} + \frac{1}{1000-N}\right) dN = \int 0.4\,dt \Rightarrow \ln N - \ln(1000 - N) = 0.4t + C.$$

(For $0 < N < 1000$ the absolute values can be dropped.) So $\dfrac{N}{1000-N} = Be^{0.4t}$ with $B = \dfrac{50}{950} = \dfrac{1}{19}$. Solving: $N = (1000 - N)\frac{1}{19}e^{0.4t}$, so

$$N(t) = \frac{1000}{1 + 19e^{-0.4t}}.$$

Check: $N(0) = 1000/20 = 50$. ✓

(b) $N = 500 \iff 19e^{-0.4t} = 1 \iff t = \dfrac{\ln 19}{0.4} = 2.5\ln 19 \approx 7.36$ years.

(c) $N' = 0.4N - 0.0004N^2$ is a downward parabola in N with top at $N = 500$. Largest growth rate: $0.4 \cdot 500 \cdot 0.5 = 100$ fish per year, at $N = 500$ (so at $t \approx 7.36$).

13. (a) $\dfrac{x^2 + 2}{3} = x \iff x^2 - 3x + 2 = 0 \iff x = 1$ or $x = 2$.

(b) $f'(x) = \frac{2}{3}x$. $f'(1) = \frac{2}{3}$: $|f'(1)| < 1$, stable (monotonic). $f'(2) = \frac{4}{3} > 1$: unstable.

(c) $u_0 = 0$: $u_1 = \frac{2}{3}, u_2 \approx 0.815, \ldots$ increasing to 1 (staircase from below). $u_0 = 1.5$: $u_1 \approx 1.417$, decreasing to 1 (staircase between 1 and 2). $u_0 = 2.5$: $u_1 = 2.75, u_2 \approx 3.19, \ldots$ increasing to ∞.

(d) Note $f(x) - x = \frac{1}{3}(x-1)(x-2)$ and f is even.

- If $|u_0| < 2$ then $u_1 = f(u_0) \in [\frac{2}{3}, 2)$. On $[0, 1)$: $f(x) > x$ and $f(x) < f(1) = 1$ (as f is increasing on $[0, \infty)$), so the sequence increases and is bounded by 1; on $(1, 2)$: $1 < f(x) < x$, so it decreases and is bounded below by 1. A monotonic bounded sequence converges, and the limit must be a fixed point (by continuity of f), so it is 1.
- If $|u_0| = 2$ then $u_1 = 2$ and the sequence stays at 2.

- If $|u_0| > 2$ then $u_1 > 2$ and $f(x) > x$ for $x > 2$, so the sequence increases; if it were bounded it would converge to a fixed point greater than 2, which does not exist. So $u_n \to \infty$.

Answer: exactly for $-2 < u_0 < 2$.

14. (a) $f(x) = 2.5x(1-x)$; fixed points 0 and $1 - \frac{1}{2.5} = 0.6$. $f'(x) = 2.5 - 5x$: $f'(0) = 2.5$, unstable; $f'(0.6) = -0.5$, stable.

(b) $f'(0.6) = -0.5 < 0$, so near 0.6 the map behaves like $u_{n+1} - 0.6 \approx -0.5(u_n - 0.6)$: the sign of $u_n - 0.6$ alternates. Oscillating convergence (a spiral in the web diagram).

15. (a) $g(y) = y(y-1)(y-3)$, equilibria $y = 0, 1, 3$. $g'(y) = 3y^2 - 8y + 3$: $g'(0) = 3 > 0$ (unstable), $g'(1) = -2 < 0$ (stable), $g'(3) = 6 > 0$ (unstable). Sign check: $g < 0$ for $y < 0$, $g > 0$ on $(0, 1)$, $g < 0$ on $(1, 3)$, $g > 0$ for $y > 3$, consistent.

(b) Horizontal lines at $y = 0, 1, 3$; between 0 and 3 curves flow towards $y = 1$; above 3 they rise steeply; below 0 they fall.

(c) $y(0) = 0.5$: increases to 1. $y(0) = 2$: decreases to 1. $y(0) = 4$: increases without bound (in fact it blows up in finite time, since $g(y) \geq cy^3$ for large y; compare problem 16).

16. $\int y^{-2}\,dy = \int dx \Rightarrow -\frac{1}{y} = x + C$; $y(0) = 1$ gives $C = -1$, so $y = \dfrac{1}{1-x}$. This is a solution on $(-\infty, 1)$: the interval containing $x = 0$ on which the formula is defined. As $x \uparrow 1$, $y \to \infty$. With $y(0) = -1$: $C = 1$, $y = -\dfrac{1}{1+x}$, which exists on $(-1, \infty)$ and tends to 0 as $x \to \infty$.

17. (a) $y_{k+1} = y_k + 0.1(-2y_k + t_k) = 0.8y_k + 0.1t_k$.

- $y_1 = 0.8 \cdot 1 + 0 = 0.8$ ($t = 0.1$)
- $y_2 = 0.64 + 0.01 = 0.65$ ($t = 0.2$)
- $y_3 = 0.52 + 0.02 = 0.54$ ($t = 0.3$)

(b) $y = \frac{1}{2}t - \frac{1}{4} + \frac{5}{4}e^{-2t}$: $y' = \frac{1}{2} - \frac{5}{2}e^{-2t}$; $-2y + t = -t + \frac{1}{2} - \frac{5}{2}e^{-2t} + t = \frac{1}{2} - \frac{5}{2}e^{-2t}$ ✓; $y(0) = -\frac{1}{4} + \frac{5}{4} = 1$ ✓. $y(0.3) = 0.15 - 0.25 + 1.25e^{-0.6} \approx 0.58601$. Error ≈ 0.0460.

(c) For example

```
def euler(f, t0, y0, h, n):
    t, y = t0, y0
    for _ in range(n):
        y = y + h * f(t, y)
        t = t + h
    return y

approx = euler(lambda t, y: -2*y + t, 0, 1, 0.01, 30)   # 0.58186
```

Error $\approx 0.58601 - 0.58186 = 0.0042$. The error shrank by a factor of about 11 when h became 10 times smaller, consistent with the error being roughly proportional to h.

18. (a) $h = 0$ is a constant solution. For $h > 0$: $\int h^{-1/2}\,dh = \int -0.1\,dt \Rightarrow 2\sqrt{h} = -0.1t + C$. $h(0) = 4$: $C = 4$. So $\sqrt{h} = 2 - 0.05t$ and $h(t) = (2 - 0.05t)^2$ for $0 \leq t \leq 40$.

(b) $h = 0$ when $t = 40$ minutes.

(c) After $t = 40$ the physical solution is $h \equiv 0$: gluing $h = (2 - 0.05t)^2$ on $[0, 40]$ to $h = 0$ for $t \geq 40$ gives a differentiable solution (both pieces have derivative 0 at $t = 40$). Now suppose all you know is that the tank is empty at $t = 50$, i.e. $h(50) = 0$. For every $t_1 \leq 50$ the function

$$h(t) = \begin{cases} (0.05(t_1 - t))^2 & t \leq t_1 \\ 0 & t \geq t_1 \end{cases}$$

solves the equation (check: for $t < t_1$, $h' = -0.005(t_1 - t) = -0.1\sqrt{h}$) and has $h(50) = 0$. So the initial value problem $h(50) = 0$ has infinitely many solutions, one for each emptying time t_1, plus $h \equiv 0$. The usual uniqueness theorem does not apply because $\sqrt{h}$ is not differentiable at $h = 0$. Physically: an empty tank does not tell you when it emptied.

Challenge

19. *Attracting case.* f' is continuous and $|f'(L)| < 1$. Choose k with $|f'(L)| < k < 1$. By continuity there is $\delta > 0$ with $|f'(x)| \le k$ for all $x \in I = [L-\delta, L+\delta]$. If $u_n \in I$, the mean value theorem gives ξ between u_n and L (so $\xi \in I$) with

$$|u_{n+1} - L| = |f(u_n) - f(L)| = |f'(\xi)|\,|u_n - L| \le k|u_n - L|.$$

So $u_{n+1} \in I$ too. By induction all $u_n \in I$ and $|u_n - L| \le k^n|u_0 - L| \le k^n\delta \to 0$.

Repelling case. Now $|f'(L)| > 1$; choose $1 < k < |f'(L)|$ and δ with $|f'(x)| \ge k$ on I. Suppose $u_0 \ne L$ and all u_n stay in I. The same argument gives $|u_n - L| \ge k^n|u_0 - L|$, which tends to infinity, contradicting $|u_n - L| \le \delta$. So the sequence must leave I.

20. (a) φ and ψ are the roots of $x^2 = x + 1$. Let $G_n = (\varphi^n - \psi^n)/\sqrt{5}$. Then $G_0 = 0$ and $G_1 = (\varphi - \psi)/\sqrt{5} = \sqrt{5}/\sqrt{5} = 1$. Strong induction: if $F_n = G_n$ and $F_{n+1} = G_{n+1}$, then

$$G_{n+1} + G_n = \frac{\varphi^n(\varphi+1) - \psi^n(\psi+1)}{\sqrt{5}} = \frac{\varphi^{n+2} - \psi^{n+2}}{\sqrt{5}} = G_{n+2},$$

using $\varphi + 1 = \varphi^2$ and $\psi + 1 = \psi^2$. So $F_{n+2} = F_{n+1} + F_n = G_{n+1} + G_n = G_{n+2}$.

(b) For $n \ge 1$: $q_{n+1} = \dfrac{F_{n+2}}{F_{n+1}} = \dfrac{F_{n+1} + F_n}{F_{n+1}} = 1 + \dfrac{1}{q_n}$. The fixed point of $f(x) = 1 + 1/x$ with $x > 0$ satisfies $x^2 = x + 1$, so it is φ; $f'(\varphi) = -1/\varphi^2 \approx -0.38$, so φ is attracting (oscillating). To get convergence for our actual starting value: all $q_n \ge 1$ (since $F_{n+1} \ge F_n$ for $n \ge 1$), and for $x \ge 1$

$$|f(x) - \varphi| = \left|\frac{1}{x} - \frac{1}{\varphi}\right| = \frac{|x - \varphi|}{x\varphi} \le \frac{|x - \varphi|}{\varphi}.$$

(We used $f(\varphi) = \varphi$, i.e. $1 + 1/\varphi = \varphi$.) Hence $|q_n - \varphi| \le \varphi^{-(n-1)}|q_1 - \varphi| \to 0$. Alternatively from (a): $q_n = \dfrac{\varphi^{n+1} - \psi^{n+1}}{\varphi^n - \psi^n} \to \varphi$ since $|\psi/\varphi| < 1$.

21. AM–GM for the $n + 1$ positive numbers n times $1 + \frac{1}{n}$ and once 1:

$$\sqrt[n+1]{\left(1 + \tfrac{1}{n}\right)^n \cdot 1} \le \frac{n\left(1 + \frac{1}{n}\right) + 1}{n+1} = \frac{n+2}{n+1} = 1 + \frac{1}{n+1}.$$

Raise to the power $n + 1$: $a_n \le a_{n+1}$. Equality in AM–GM holds only if all numbers are equal, which they are not, so $a_n < a_{n+1}$.

22. (a) Write $f(f(x)) - x$. It vanishes at the fixed points $x = 0$ and $x = 1 - 1/r$, so $x(rx - r + 1)$ divides it. Expanding and dividing (or using SymPy) gives

$$f(f(x)) - x = -x\,(rx - r + 1)\,\left(r^2x^2 - r(r+1)x + (r+1)\right).$$

So the points of a genuine 2-cycle satisfy $r^2x^2 - r(r+1)x + (r+1) = 0$.

(b) Discriminant: $r^2(r+1)^2 - 4r^2(r+1) = r^2(r+1)(r-3)$. For $r > 0$ this is positive iff $r > 3$. (At $r = 3$ the double root is $x = \frac{2}{3}$, the fixed point itself.) So a 2-cycle exists exactly for $r > 3$, which is exactly where the fixed point $1 - \frac{1}{r}$ became unstable: a *period-doubling bifurcation*.

(c) $x = \dfrac{(r+1) \pm \sqrt{(r+1)(r-3)}}{2r} = \dfrac{4.2 \pm \sqrt{0.84}}{6.4}$, so $x_1 \approx 0.7995$, $x_2 \approx 0.5130$. Check: $3.2 \cdot 0.7995 \cdot 0.2005 \approx 0.5130$ and $3.2 \cdot 0.5130 \cdot 0.4870 \approx 0.7995$. ✓

23. $(ye^{2t})' = y'e^{2t} + 2ye^{2t} = (y' + 2y)e^{2t} = e^t e^{2t} = e^{3t}$. Integrate: $ye^{2t} = \frac{1}{3}e^{3t} + C$, so

$$y = \tfrac{1}{3}e^t + Ce^{-2t}.$$

Check: $y' + 2y = \frac{1}{3}e^t - 2Ce^{-2t} + \frac{2}{3}e^t + 2Ce^{-2t} = e^t$. ✓

24. Write $g(N) = rN(1 - N/K)$. Since $0 < N(0) < K$ and $N = 0$, $N = K$ are solutions, the solution stays strictly between 0 and K (two solution curves of this equation cannot cross, by uniqueness), so $N' = g(N) > 0$ for all t: N is increasing. By the chain rule

$$N'' = g'(N)\,N' = r\left(1 - \frac{2N}{K}\right)N'.$$

Because $N' > 0$, the sign of N'' is the sign of $1 - 2N/K$: positive while $N < K/2$, negative when $N > K/2$. Since N starts below $K/2$, increases, and tends to K (stable equilibrium), it passes $K/2$ exactly once (it is strictly increasing). So the graph is convex then concave with exactly one inflection point, at height $K/2$. The slope there is $g(K/2) = r \cdot \frac{K}{2} \cdot \frac{1}{2} = \frac{rK}{4}$.